Question-1

What is the main goal of using an ensemble model in machine learning?	
Option A	To increase the speed of training
Option B	To combine the predictions of multiple models to improve accuracy
Option C	To reduce the size of the dataset.
Option D	To remove the need for data preprocessing
Solution	
Correct Option/s	B: To combine the predictions of multiple models to improve accuracy
Solution Explanation	Correct, because ensemble models combine multiple models to leverage their individual strengths, leading to better overall accuracy and reduced errors, much like getting advice from multiple friends. **Option A: To increase the speed of training** **Incorrect**, because using multiple models in an ensemble generally slows down training as it involves training many models rather than speeding up the process. **Option C: To reduce the size of the dataset** **Incorrect**, because ensemble models do not alter the dataset size. They focus on combining predictions from models, not on changing the data. **Option D: To remove the need for data preprocessing** **Incorrect**, because ensemble models still require proper data preprocessing. They don't eliminate the need for good-quality input data.

Question-2

Why would we use a random forest instead of a decision tree?	
Option A	For lower training error.
Option B	To reduce the underfitting problem.

Option C	For a model that is easier for a human to interpret.
Option D	To reduce the variance of the model.
Solution	
Correct Option/s	D
Solution Explanation	Random forests have lower variance compared to decision trees. Option A suggests that we would use a random forest instead of a decision tree for lower training error. While random forests often have lower training error compared to individual decision trees due to their ensemble nature, this is not the primary reason for choosing random forests over decision trees. Option B suggests that we would use a random forest to reduce the underfitting problem. While random forests can help mitigate underfitting compared to single decision trees, they are primarily used to address overfitting, not underfitting. Option C is incorrect because random forests are generally more complex and less interpretable compared to individual decision trees, so they are not chosen for their interpretability.

Question-3

In bagging, each individual model in the ensemble is trained on:	
Option A	The entire training dataset
Option B	A different random subset of features
Option C	A different random subset of data points (with replacement)
Option D	A different random subset of data points (without replacement)
Solution	
Correct Option/s	C: A different random subset of data points (with replacement)
Solution Explanation	Option C: Correct. Bagging uses bootstrapped datasets created by sampling data points with replacement. This introduces diversity among the models, which reduces variance and helps in preventing overfitting. Option A: Incorrect. Training on the entire dataset would make the models highly correlated, defeating the purpose of bagging,

	which is to create diversity among models. Option B: Incorrect. Feature bagging involves subsets of features, not data points. Bagging primarily focuses on subsets of data points. Option D: Incorrect. Sampling without replacement would produce identical datasets in the ensemble, eliminating the randomness that bagging requires to reduce variance.

Question-4

Why might a Random Forest be less interpretable than a single decision tree?	
Option A	Because it uses a larger number of features.
Option B	Because it combines the predictions of many decision trees, making it hard to trace the decision path.
Option C	Because it requires more data preprocessing.
Option D	Because it uses complex mathematical functions for prediction.
Solution	
Correct Option/s	B: Because it combines the predictions of many decision trees, making it hard to trace the decision path.
Solution Explanation	**Option B:** Correct. A Random Forest is an ensemble method that aggregates the predictions of many individual decision trees. While each tree in the forest is interpretable on its own, the final prediction is made by combining the outputs of all the trees, typically through majority voting (for classification) or averaging (for regression). This aggregation makes it difficult to trace the exact decision path that led to the final prediction, reducing the model's interpretability compared to a single decision tree, where you can easily follow the decision-making process from root to leaf. **Option A:** Incorrect. The number of features used doesn't directly impact interpretability. Even if a Random Forest uses a large number of features, it's the ensemble nature of combining multiple trees that complicates interpretation. **Option C:** Incorrect. Random Forests do not inherently require more data preprocessing than a single decision tree. In fact, they often require less feature engineering since they can handle unprocessed data relatively well. **Option D:** Incorrect. Random Forests use decision trees, which are not based on complex mathematical functions but on simple

	if-else conditions. The complexity arises from combining multiple trees, not from the use of intricate functions.

Question-5 (Multi-Correct)

If we are training a decision tree and the model has a high training error, what can we do to fix this problem? (Multi-Correct)	
Option A	Increase minimum samples at which the tree stops splitting.
Option B	Decrease minimum samples at which the tree stops splitting.
Option C	Increase tree depth
Option D	Reduce tree depth
Solution	
Correct Option/s	B, C
Solution Explanation	If the model training error is high, it means the model has a high bias/underfitting problem. Increasing tree depth/ reducing min samples to stop splitting improves model complexity, which can solve the underfitting problem. A is incorrect because increasing the minimum samples at which the tree stops splitting (also known as increasing the minimum samples per leaf) typically helps prevent overfitting rather than directly addressing high training errors. D is incorrect because reducing tree depth typically helps prevent overfitting by limiting the complexity of the model, but it may not directly address high training error if the underlying patterns in the data require a deeper tree to be captured accurately.

Question-6

In a Random Forest, which is an example of bagging, how is the final prediction made for classification tasks?	
Option A	By averaging the predictions of all individual trees
Option B	By selecting the prediction with the highest confidence score
Option C	By majority voting from all individual trees

Option D	By taking the median of the predictions from all individual trees
Solution	
Correct Option/s	C: By majority voting from all individual trees
Solution Explanation	**Option C:** Correct. In Random Forest, the final classification prediction is made by majority voting, where the class that receives the most votes from the individual trees is chosen as the final output. **Option A:** Incorrect. Averaging is used in regression tasks, not classification. **Option B:** Incorrect. Confidence scores are not typically used in Random Forest; instead, predictions are based on votes. **Option D:** Incorrect. Taking the median applies to regression predictions, not classification.

Question-7

Suppose you train a different variant of the random forest model where instead of choosing a random subset of features you specify a constraint that a feature can be only used in K splits in a tree. Which of the following statement is true for the resulting ensemble model	
Option A	The variance of the ensemble model increases
Option B	The bias of the final ensemble increases.
Option C	The bias of the final ensemble decreases.
Option D	None of the above
Solution	
Correct Option/s	B
Solution Explanation	If we use a non-random feature selection criteria, we are introducing a bias hence the bias of the final ensemble increases..

Question-8 (Multi-Correct)

Which among the following are some of the differences between bagging and boosting? (Multi-Correct)	
Option A	In bagging we use the same classification algorithm for training on each sample of the data, whereas in boosting, we use different classification algorithms on the different training data samples
Option B	Bagging is easy to parallelize whereas boosting is inherently a sequential process
Option C	In bagging we typically use sampling with replacement whereas in boosting, we typically use weighted sampling techniques
Option D	In comparison with the performance of a base classifier on a particular data set, bagging will generally not increase the error whereas as boosting may lead to an increase in the test error
Solution	
Correct Option/s	B, C, D
Solution Explanation	B) Bagging (Bootstrap Aggregation) is an ensemble learning method which trains multiple models independently in parallel. Boosting is an ensemble learning method which trains each new model such that it focuses on correcting the errors made by the previous model. C) In the case of Bagging, any element has the same probability of appearing in a new data set. Training data subsets are drawn randomly with a replacement for the training dataset. However, for Boosting, the observations are weighted. In Boosting algorithms, each classifier is trained on data, considering the previous classifiers' success. Hence, every new training subset comprises the elements misclassified by previous models. Misclassified data increases its weight to emphasize the most difficult cases. D) Boosting can result in an increase in test error over a base classifier due to over-emphasis on existing noise data points in later iterations.

Question-9

In Random forest we can generate hundreds of trees (say T1, T2 …..Tn) and then

aggregate the results of these trees. Which of the following is true about the individual (T_k) tree in Random Forest?

1. Individual tree is built on a subset of the features
2. Individual tree is built on all the features
3. Individual tree is built on a subset of observations
4. Individual tree is built on a full set of observations

Option A	1 and 3
Option B	1 and 4
Option C	2 and 3
Option D	2 and 4
Solution	
Correct Option/s	A
Solution Explanation	Random forest is based on the bagging concept, that considers a fraction of samples and a fraction of features for building the individual trees. Option B is incorrect because while it correctly states that each tree is built on a subset of the features, it incorrectly suggests that each tree is built on a full set of observations. Option C is incorrect because it states that each tree is built on all the features, which is not true for Random Forest. Option D is incorrect because it suggests that each tree uses all features and all observations, which contradicts the fundamental operation of Random Forest.

Question-10

Suppose we are given the following scenarios for training and validation errors for Gradient Boosting. Which of the following hyperparameters should be chosen in such a case?

Scenario	Depth	Training Error	Validation Error
1	2	100	110

2	4	90	105
3	6	50	120
4	8	45	105
5	10	30	150

Option A	Scenario - 2
Option B	Scenario - 3
Option C	Scenario - 1
Option D	Scenario - 4
Solution	
Correct Option/s	A
Solution Explanation	Scenario 2 and 4 have the same validation accuracies, but we would select 2 because depth is lower is a better hyperparameter. Also the training validation loss difference is smaller which is an indicator of good generalization. Scenario 1 indicates underfitting, where the model is too simple to capture the underlying data structure effectively. Scenarios 3, 4, and 5 show signs of overfitting as the model depth increases, evidenced by the increasing discrepancy between training and validation errors.

Question-11

For an ensemble method to perform better, which of the following properties the individual base models should have Note: Assume each base model has an accuracy greater than 50%.	
Option A	Less correlation among predictions
Option B	High correlation among predictions
Option C	Correlation does not play any role in final ensemble model performance
Option D	None of These

Solution	
Correct Option/s	A
Solution Explanation	Ensemble models perform well when models are less correlated. Option B is incorrect because if the individual models are highly correlated in their predictions, they are likely capturing the same patterns or errors, and thus, the ensemble might not perform significantly better than any single model. Essentially, the ensemble fails to reduce the model-specific errors effectively, limiting its potential to outperform its components. Option C is incorrect because correlation among predictions plays a crucial role in the effectiveness of an ensemble. Low correlation among model predictions typically enhances the ensemble's performance by aggregating diverse perspectives on the data, thereby improving robustness and accuracy.

Question-12

Suppose, we are working on a binary classification problem. And there are 3 models, each with 70% accuracy. If we want to ensemble these models using the majority voting method. What will be the maximum and minimum accuracy we can get, respectively?	
Option A	100%, less than 70%
Option B	100%, more than 70%
Option C	less than 100%, 70%
Option D	Less than 70%, 100%
Solution	
Correct Option/s	A: 100%, less than 70%
Solution Explanation	**Maximum Accuracy (100%):** If all three models agree on the correct class for each instance, the ensemble's accuracy will be 100%. This is because majority voting would always result in the correct class being chosen when there is unanimous agreement. **Minimum Accuracy (less than 70%):** In some cases, the ensemble accuracy can drop below the accuracy of the individual models. This happens when the models disagree on the majority class frequently, leading to incorrect majority votes. For example, if the two out of the three models are wrong on a particular instance, the majority vote will also be wrong, leading to an accuracy lower than that of the individual models. We illustrate this using two examples. 1. For maximum accuracy, ensemble accuracy is 100%

Actual output	M1	M2	M3	Ensemble Output
1	1	0	1	1
1	1	0	1	1
1	1	0	1	1
1	0	1	1	1
1	0	1	1	1
1	0	1	1	1
1	1	1	1	1
1	1	1	0	1
1	1	1	0	1
1	1	1	0	1

2. For minimum accuracy, ensemble accuracy is 60%

Actual output	M1	M2	M3	Ensemble Output
1	1	0	0	0
1	1	1	1	1
1	1	0	0	0
1	0	1	0	0
1	0	1	1	1
1	0	0	1	0
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

Question-13

For the ‘max_depth’ hyperparameter used in Gradient Boosting algorithm, which of the following statements can be made?

1. Lower is the better parameter in case of same validation accuracy
2. Higher is the better parameter in case of same validation accuracy
3. Increasing the value of max_depth may overfit the data
4. Increasing the value of max_depth may underfit the data

Option A	1 and 3
Option B	1 and 4
Option C	2 and 3
Option D	2 and 4
Solution	

Correct Option/s	A
Solution Explanation	**Lower is the better parameter in case of same validation accuracy:** When two models achieve the same validation accuracy, the model with the lower complexity (i.e., lower `max_depth`) is typically preferred. This is because a simpler model is less likely to overfit and usually generalizes better to new, unseen data. Overfitting occurs when a model becomes too complex and starts capturing noise in the training data, leading to poor performance on new data. **Increasing the value of `max_depth` may overfit the data:** A higher `max_depth` allows the decision trees to capture more details and complexities in the training data, which can lead to overfitting, especially if the depth is too high relative to the complexity of the data. Overfitting happens when the model fits the training data very well but fails to generalize to new data. **Higher is the better parameter in case of same validation accuracy:** This is incorrect. Increasing the `max_depth`unnecessarily, without an increase in validation accuracy, adds complexity to the model, which could lead to overfitting without any performance benefit. **Increasing the value of `max_depth` may underfit the data:** This is generally false. Underfitting occurs when the model is too simple (e.g., `max_depth` is too low) to capture the underlying patterns in the data, not when the model is complex.

Question-14

A Gradient Boosting model is trained on a dataset of millions of observations and thousands of features. Before building the model, let us consider different parameter settings for time measurement. Consider the hyperparameter “number of trees” and arrange the options in terms of time taken by each hyperparameter for building the Gradient Boosting model? 1. Number of trees = 100 2. Number of trees = 500 3. Number of trees = 1000	
Option A	1~2~3
Option B	1 < 2 < 3
Option C	3 < 2 < 1
Option D	None of These
Solution	
Correct Option/s	B

Solution Explanation	● **Time Complexity in Gradient Boosting:** The number of trees in a Gradient Boosting model directly impacts the time it takes to train the model. Gradient Boosting builds trees sequentially, where each tree is trained to correct the errors made by the previous trees, necessitating a sequential training process.. This sequential nature means that as the number of trees increases, the time required to train the model also increases. ● **Option B (1 < 2 < 3) is Correct:** This option correctly arranges the number of trees in ascending order, reflecting the increasing time complexity. For instance, training 1000 trees will take more time than training 100 trees, because each tree is built one after the other. ● **Option C (3 < 2 < 1) is Incorrect:** This option incorrectly suggests that increasing the number of trees would decrease the training time, which is contrary to the actual process where more trees require more time. In Gradient Boosting, you cannot parallelize the training of trees since each tree depends on the outcome of the previous one, leading to a linear increase in time with the number of trees.

Question-15

What is a unique feature of XGBoost that enhances its accuracy compared to traditional Gradient Boosting?	
Option A	XGBoost employs a linear booster in addition to tree boosters
Option B	XGBoost uses logistic regression as its default weak learner, while Gradient Boosting uses decision trees.
Option C	XGBoost provides tools for efficient hyperparameter tuning, while Gradient Boosting does not.
Option D	XGBoost includes both L1 and L2 regularization terms to control model complexity.
Solution	
Correct Option/s	D: XGBoost includes both L1 and L2 regularization terms to

	control model complexity.
Solution Explanation	**Option D: Correct:** XGBoost incorporates L1 (Lasso) and L2 (Ridge) regularization, which helps in controlling the complexity of the model, reducing overfitting, and improving accuracy. This feature is not present in traditional Gradient Boosting. **Option A: Incorrect:** While XGBoost can use a linear booster, its primary and most commonly used booster is still decision trees, similar to traditional Gradient Boosting. **Option B: Incorrect:** Both XGBoost and traditional Gradient Boosting primarily use decision trees as weak learners. XGBoost does not use logistic regression by default. **Option C: Incorrect:** While XGBoost provides tools for efficient hyperparameter tuning, such as cross-validation and early stopping, these tools do not automatically select the best hyperparameters without user intervention. Hyperparameters still need to be tuned by the user, often through processes like grid search or Bayesian optimization. Traditional Gradient Boosting methods can also involve hyperparameter tuning, though XGBoost offers more advanced options for this purpose